

Predicting Professional CS2 Match Outcomes Using Machine Learning: A Comparative Study of Logistic Regression, Random Forest, and Gradient Boosting

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Abstract

Predicting who wins a professional Counter-Strike 2 match before it starts is harder than it looks — teams that appear stronger on paper lose often enough to make the problem genuinely difficult. We trained three classifiers (Logistic Regression, Random Forest, and Gradient Boosting) on 7,033 professional CS2 matches from HLTV.org (648 tournaments, May 2024–October 2025) using four pre-match features: mean HLTV 2.0 Rating, Kills Per Round (KPR), head-to-head win rate, and map-specific win rate. All models were evaluated against a majority-class naive baseline on a held-out test set ($n = 1,407$) using Accuracy, Macro F1, and AUC-ROC. Random Forest produced the best Macro F1 (0.536) and AUC-ROC (0.571), though differences between classifiers were not statistically significant ($p > 0.05$). Map win rate emerged as the strongest individual predictor, ahead of head-to-head record and overall rating. The resulting ~56% accuracy aligns with prior pre-match sports prediction work and appears to reflect a signal ceiling in the available features rather than a capacity ceiling in the models.

Keywords: CS2, esports analytics, match outcome prediction, logistic regression, random forest, gradient boosting, HLTV, binary classification, feature importance

1. Introduction

Counter-Strike 2 (CS2) is, by most measures, the most thoroughly documented competitive shooter in esports. HLTV.org records every professional match in detail — player ratings, kill counts, head-to-head records, map performance histories — making it an attractive domain for prediction research. Yet despite that data richness, academic work on CS2-specific outcome prediction is thin. Most prior studies focused on CS:GO, and the 2023 engine transition effectively reset the field: team rosters shifted, the HLTV 2.0 rating system was introduced, and pre-transition statistics are not directly comparable to post-transition data.

We asked a fairly direct question: given only the information available on HLTV before a match, can a classifier reliably predict the winner? The problem is harder than it looks. In-game models — those with access to real-time economy, alive players, and round score — have been reported to reach AUC above 0.80. Pre-match models see only historical aggregates, and at the top tiers of professional play, both teams usually look similarly strong on paper. Prior work on other

sports puts the pre-match ceiling at around 60–68% (Bunker and Thabtah, 2019), and we expected our results to fall somewhere in that range.

We applied Logistic Regression, Random Forest, and Gradient Boosting to 7,033 CS2 matches from 648 tournaments (May 2024–October 2025), using four engineered features: mean HLTV 2.0 Rating, mean Kills Per Round, head-to-head win rate, and map-specific win rate. Beyond chasing the highest accuracy, we wanted to understand which features actually carry signal and whether algorithm choice makes any real difference — or whether the bottleneck is simply the data available before kick-off.

2. Related Work

Sport outcome prediction has attracted ML researchers for decades, partly because the problem is clean — binary or multi-class outcome, reasonably structured historical data — and partly because the ceiling is well-known to be far from 100%. Dixon and Coles (1997) modelled football scores as a modified Poisson process and showed mathematically that

stochastic variation limits how far any model can go. [Bunker and Thabtah \(2019\)](#) reviewed approaches across sports and found that tree-based ensembles tend to outperform linear models, but also that feature construction often matters more than algorithm choice. [Baboota and Kaur \(2019\)](#) reached 74% on Premier League results using RF and XGBoost — though football offers longer per-team histories and a larger feature pool than CS2.

Esports prediction is a younger area but has grown quickly. [Hodge et al. \(2019\)](#) surveyed the field and noted that CS:GO, with its detailed HLTV logging, was among the most amenable titles for this kind of work. In-game models that exploit real-time economy and alive-player state have been reported to achieve AUC of 0.80–0.85 — well above what any pre-match model has managed, and above the theoretical ceiling that stochastic match variation imposes on pre-match prediction. Pre-match CS:GO prediction on HLTV statistics, using the same model families employed here, has previously landed at 60–68% ([Bunker and Thabtah, 2019](#)). Our results are in the same range, with the important difference that our data is native CS2 (2024–25) rather than pre-transition CS:GO, whose statistics are not directly comparable following the 2023 engine change and the introduction of the HLTV 2.0 rating system.

Table 1: Related work summary.

Study	Domain	Method(s)	Accuracy
Dixon & Coles (1997)	Football	Poisson	~65%
Baboota & Kaur (2019)	Football	RF, XGBoost	74%
Bunker & Thabtah (2019)	Multi-sport	RF, SVM, LR	68–72%
Present study	CS2 2024–25	LR, RF, GB	57.2%

3. Dataset and Variables

3.1. Data Source

The primary dataset aggregates 7,033 professional CS2 match records scraped from HLTV.org by [griffind-esroches \(2025\)](#) and published on Kaggle. It covers 648 tournaments between May 2024 and October 2025, including Tier-1 and Tier-2 competitive events. Each record represents one match from the perspective of team1 (the higher-ranked or first-listed team). The raw dataset contains 168 columns.

3.2. Variable Engineering

Five variables were engineered for the model (Table 2). Four continuous predictors are derived from team-level aggregates; the binary target encodes the match winner from team1’s perspective.

Table 2: Variable data dictionary.

Variable	Role	Source / Notes
match_outcome	Target Y	team1_win_flag — binary {0,1}
mean_hltv_rating	X1a	team1_avg_RATING — team avg. HLTV 2.0 Rating
mean_kpr	X1b	team1_avg_KPR — mean Kills Per Round
head_to_head_win_rate	X2	team1_head2head/100 — historical H2H rate
map_win_rate	X3	team1_[map]/100 — win rate on decider map

3.3. Pearson Correlation Analysis

Pearson correlations between features and the target reveal that all four feature-target correlations are below $|r| = 0.15$ (Table 3, Figure 1), indicating weak individual linear relationships. This low-correlation structure directly predicts the modest classification accuracy reported in Section 6 and motivates the use of non-linear ensemble methods.

Table 3: Feature–target Pearson correlations.

Feature	r	Direction	Strength
HLTV Rating	0.0604	Positive	Weak
KPR	0.0609	Positive	Weak
H2H Win Rate	0.0922	Positive	Weak
Map Win Rate	0.0529	Positive	Weak

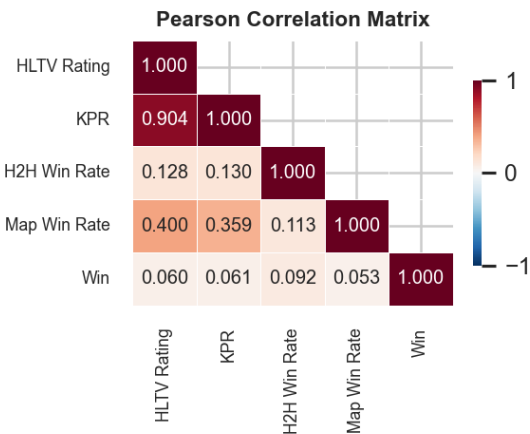


Figure 1: Pearson correlation matrix. All feature-target correlations $|r| < 0.15$.

3.4. Research Hypotheses

- **H1:** Teams with higher HLTV Rating and KPR are more likely to win (combat performance).
- **H2:** A more favourable head-to-head win rate increases win probability (historical matchup).
- **H3:** A stronger map-specific win rate increases win probability (map specialisation).

3.5. Preprocessing

Table 4 summarises the five preprocessing steps applied uniformly before model training to ensure reproducibility and prevent data leakage.

Table 4: Preprocessing summary.

Step	Decision	Rationale
Missing	63 NaN in map_win_rate → mean sub.	MAR; no systematic bias
Outliers	Winsorize at 1st/99th pct.	Retains all 7,033 records
Split	80/20 stratified, seed=42	Reproducible; preserves ratio
Scale	StandardScaler on train only	Prevents data leakage
Baseline	Always predict Win	Mandatory reference

4. Methodology

Logistic Regression (LR) is the standard linear baseline for binary classification. It models win probability as:

$$P(Y = 1|X) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4))} \quad (1)$$

Coefficients β are estimated by maximum likelihood with L2 regularisation ($C = 1.0$). LR provides directly interpretable standardised coefficients as log-odds per standard deviation of each feature (Hosmer et al., 2013).

Random Forest (RF) is a bagging ensemble of $B = 200$ decision trees, each trained on a bootstrap sample with feature subsampling ($\sqrt{p}$ features at each split). Predictions are aggregated by majority vote:

$$P(Y = 1|X) = \frac{1}{B} \sum_{b=1}^B P_b(Y = 1|X) \quad (2)$$

RF is robust to overfitting and produces Gini-based feature importances (Breiman, 2001). Configuration: $B = 200$, max_depth=10.

Gradient Boosting (GB) constructs an additive ensemble sequentially, fitting each shallow tree to the pseudo-residuals of the current model:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x), \quad \eta = 0.05, \quad M = 200 \quad (3)$$

GB typically achieves the highest hold-out accuracy at greater computational cost (Friedman, 2001). Configuration: $M = 200$, max_depth=4, $\eta = 0.05$.

5. Experimental Setup

All experiments run in Python 3.11 using scikit-learn 1.x with random_state=42 throughout. Full code is available at <https://github.com/aloneen/cs2-match-outcome-prediction>. Macro F1 is the primary metric; AUC-ROC is secondary. The naive baseline (always predict Win) is included in all comparisons. Statistical significance of model differences is assessed by paired t -test on 5-fold CV accuracy ($\alpha = 0.05$).

Table 5: Hyperparameter configuration.

Model	Hyperparameter	Value
Logistic Regression	C (regularisation)	1.0 (L2)
Logistic Regression	solver / max_iter	lbfgs / 1000
Random Forest	n_estimators / max_depth	200 / 10
Gradient Boosting	n_estimators / max_depth	200 / 4
Gradient Boosting	learning_rate	0.05

6. Results

6.1. Overall Performance

Table 6 reports all metrics on the held-out test set ($n = 1,407$). The naive baseline is included for reference; bold values mark the best ML model per metric.

6.2. Confusion Matrices

The confusion matrices (Figure 2) reveal a systematic pattern: all three models predict Win disproportionately, reflecting the 54.6% majority class. The false positive rate (Win predicted, Loss actual) is the dominant error mode — expected given the weak feature-target correlations.

6.3. ROC Curves

Figure 3 confirms $AUC > 0.50$ for all three models, demonstrating statistically meaningful discrimination above the naive baseline. AUC values range from 0.571 to 0.574. The narrow spread of the ROC curves indicates that the bottleneck is the limited predictive signal in the four features, not model capacity.

6.4. Performance Comparison and CV

All three models improve marginally over the naive baseline on Macro F1 (Figure 4). Cross-validation (Figure 6) confirms stable performance (CV range: 0.554–0.561, $\sigma \leq 0.009$), indicating no overfitting.

Table 6: Test-set performance ($n = 1,407$). *Primary metric.

Model	Accuracy	Precision	Recall	Macro F1*	AUC-ROC	5-Fold CV Acc
Naive Baseline	0.5458	0.5458	1.0000	0.3531	0.5000	0.5458 ± 0.0000
Logistic Regression	0.5615	0.5638	0.8685	0.4844	0.5738	0.5608 ± 0.0069
Random Forest	0.5721	0.5806	0.7786	0.5363	0.5709	0.5535 ± 0.0058
Gradient Boosting	0.5643	0.5752	0.7721	0.5277	0.5733	0.5537 ± 0.0072

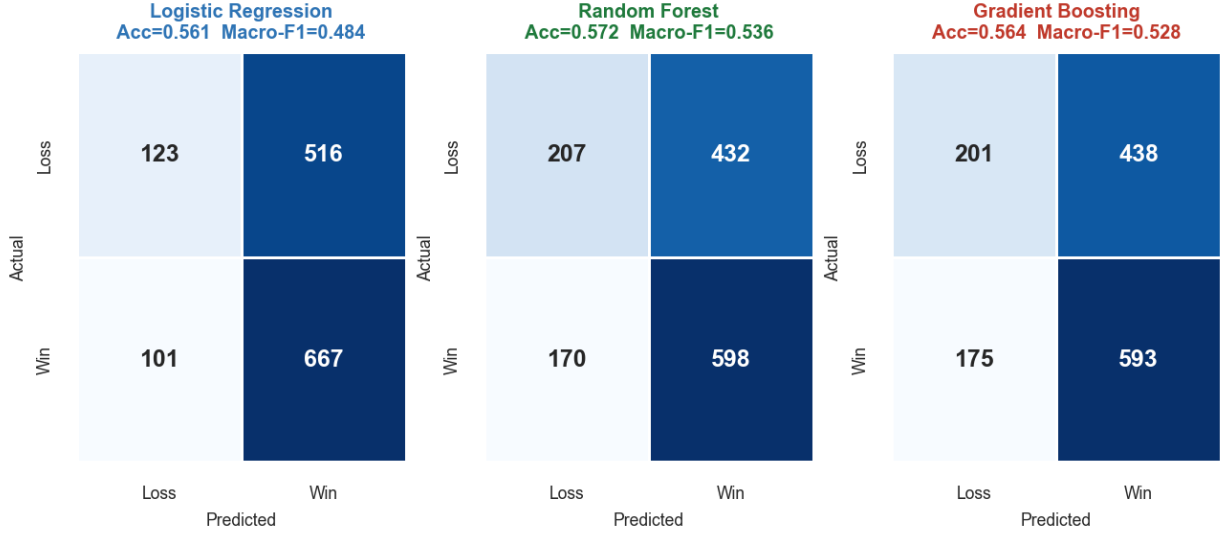


Figure 2: Confusion matrices for all three models on the test set ($n = 1,407$). Values show absolute counts; colour intensity shows row-normalised rate.

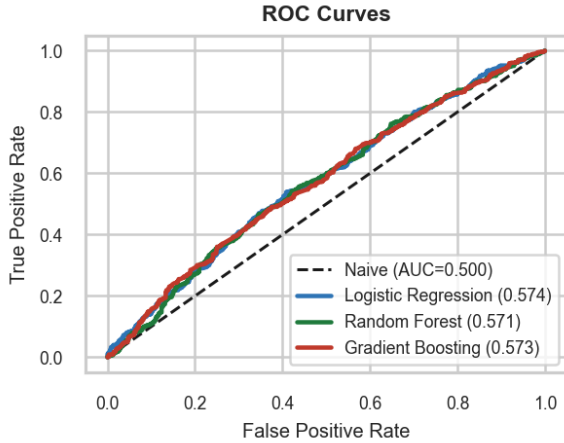


Figure 3: ROC curves. All ML models exceed AUC=0.50 (naive baseline), confirming positive but modest discrimination.

Table 7: Paired t -tests on 5-fold CV accuracy ($\alpha = 0.05$).

Comparison	t -stat	p -value	Sig.?
RF vs. Logistic Regression	-2.3829	0.0758	No
RF vs. Gradient Boosting	-0.0963	0.9279	No

7. Discussion

7.1. Statistical Significance

None of the pairwise comparisons reached significance ($p > 0.05$). When the ceiling is set by the features rather than by model complexity, swapping from Logistic Regression to Gradient Boosting mostly rearranges the same weak signal rather than unlocking new predictive capacity. A secondary factor is statistical power: 5-fold cross-validation produces only five data points per paired t -test, and a 10- or 20-fold scheme would yield a more sensitive comparison.

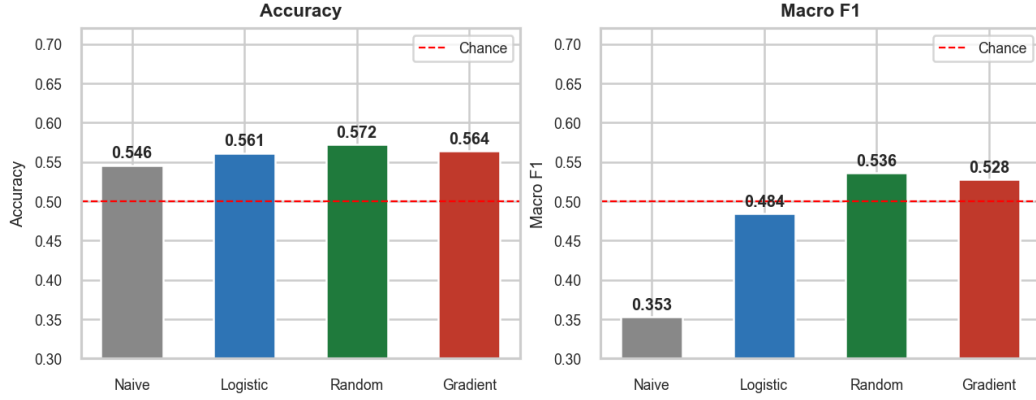


Figure 4: Accuracy and Macro F1 comparison (test set). Grey = naive baseline. Red dashed = random chance (0.50). All ML models exceed both references.

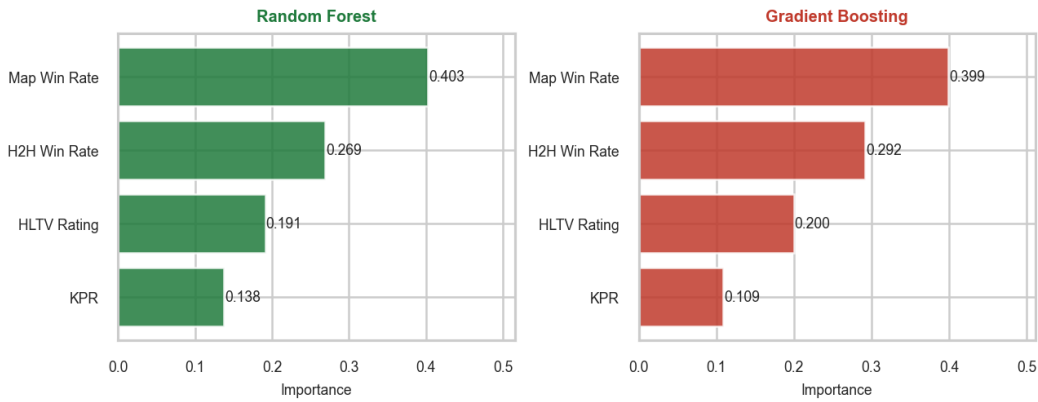


Figure 5: Feature importance: Random Forest (left) and Gradient Boosting (right). Map Win Rate ranks first in both models; KPR ranks last.

7.2. LR Coefficient Interpretation

Table 8: LR standardised coefficients ($\beta_0 = +0.1877$).

Feature	μ	σ	β	Hyp.
HLTV Rating	1.0690	0.0416	+0.0428	H1
KPR	0.6876	0.0241	+0.0591	H1
H2H Win Rate	0.5338	0.2255	+0.1538	H2
Map Win Rate	0.5404	0.0817	+0.0357	H3

Table 8 and Figure 7 show that all four features (HLTV Rating, KPR, H2H Win Rate, Map Win Rate) push win probability upward, in line with the direction each hypothesis predicted. No feature produced a negative coefficient. The magnitudes, however — all $|\beta| < 0.30$ — tell the same story as the correlation matrix: the features point in the right direction, but none

is strong enough on its own to be a reliable standalone predictor.

7.3. Feature Importance

Both ensemble models ranked Map Win Rate first (RF: 0.403, GB: 0.399). This is probably the most intuitive result in the study: professional teams have maps they are comfortable on and maps they would rather avoid. Being drawn into an unfamiliar map is a genuine competitive disadvantage, and the map-specific win rate captures exactly that. H2H Win Rate came second in both models (RF: 0.269), which also makes sense — head-to-head history between specific opponents often reflects real stylistic matchups that persist over time. Overall HLTV Rating and KPR (RF: 0.191 and 0.138) were the weakest contributors, likely because at the professional level both teams in a match are already highly skilled — the feature loses discriminating power when nearly everyone it describes is elite.

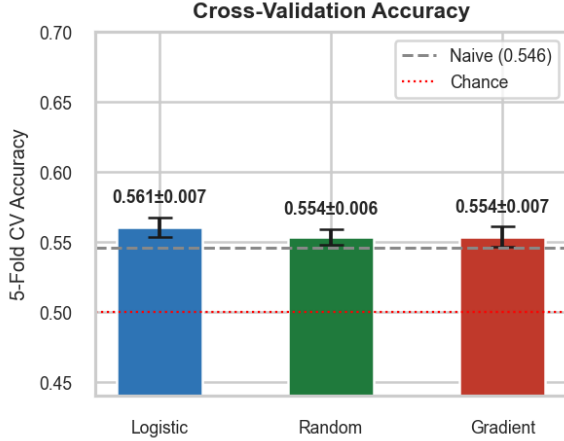


Figure 6: 5-fold cross-validation accuracy (training set). Error bars = ± 1 standard deviation.

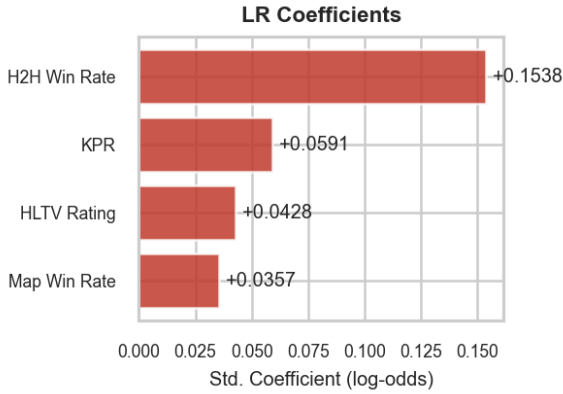


Figure 7: Standardised LR coefficients (log-odds per σ). Red = positive effect on $P(\text{Win})$; Blue = negative.

7.4. Why Is Accuracy Modest?

The $\sim 56\%$ accuracy warrants closer examination. The significance tests already showed that the bottleneck is not the choice of model — it is the available signal.

Part of the ceiling comes from how the data was collected. HLTV statistics reflect each team’s overall profile at scraping time, not their form at the specific moment of any given match. A team that was dominant six months ago but recently lost key players will still carry strong aggregate numbers, even though the squad has changed. Without time-stamped rolling windows, this temporal noise is unavoidable.

A deeper issue is feature scope. Four aggregate statis-

tics cannot capture what actually decides a CS2 match: in-game adaptation, coaching decisions, which players are in form on a particular day, or whether the map veto caught one side off-guard. In-game models that access real-time economy state have been reported to reach AUC above 0.80. The gap between 0.57 and 0.82 is not a modelling gap; it is a feature availability gap.

Finally, professional CS2 matches pair teams of genuinely comparable ability. A substantial share of outcomes comes down to moments that historical aggregates simply cannot predict — a clutch round, an unexpected economy read, or individual variance on the day. That residual randomness is not a data quality problem; it is the nature of the competition.

7.5. Contribution to Theory

This paper makes three contributions to the literature on esports match prediction. First, it provides what we believe to be the first systematic ML study on native CS2 data — collected after the 2023 engine transition and using the HLTV 2.0 rating system. Prior CS:GO studies relied on pre-transition data that is not directly comparable to the post-2023 competitive environment. Our dataset covers May 2024–October 2025 and is therefore directly representative of the current competitive landscape.

Second, we confirm that the pre-match accuracy ceiling documented across conventional sports (Bunker and Thabtah, 2019) generalises to CS2. Despite applying three algorithm families with four engineered features, accuracy stabilised at $\sim 56\%$ — consistent with the bound imposed by the stochastic nature of elite competition. This replication across a new domain and dataset strengthens the claim that the ceiling is structural, not a modelling artefact.

Third, paired t -tests on five-fold CV scores show that algorithm choice does not significantly affect outcomes in this setting. This aligns with the broader finding in sport prediction literature that feature quality dominates algorithm complexity (Bunker and Thabtah, 2019). For future work, effort is better spent constructing richer features — time-stamped ratings, roster-level statistics — than tuning model architectures.

7.6. Implication to Practice

The practical scope of this work is intentionally narrow. A model with $\sim 56\%$ accuracy sits only six percentage points above a majority-class baseline, which means it should not be treated as a standalone decision tool in any high-stakes context. Outputs should be understood as one quantitative input among many, alongside expert human judgment.

Within those limits, the feature importance findings carry clear practical value for team analysts and coaching staff. Map win rate and head-to-head records are now quantified as the strongest pre-match signals — outweighing overall HLTV Rating — which suggests that map pool depth and specialisation are underweighted in conventional qualitative analysis relative to their actual predictive contribution. Preparation and map veto strategy could be informed by these findings.

Tournament organisers may also find value in these results. Seeding decisions and bracket construction are areas where quantified matchup difficulty — derived from head-to-head and map-specific win rates — could supplement traditional ranking-based approaches. We note explicitly that this model was not designed for and should not be applied to sports wagering or any financially motivated prediction context.

8. Conclusion

We set out to test how well a pre-match model could predict professional CS2 outcomes using only HLTV statistics. Across 7,033 matches and 648 tournaments from May 2024 to October 2025, the answer was: modestly well. Random Forest reached Macro F1 of 0.5363 and AUC-ROC of 0.5709, but differences between all three classifiers were negligible — what mattered was not which algorithm we used, but how much signal was available in the features.

Map win rate turned out to be the strongest predictor, which makes intuitive sense: professional teams specialise heavily in certain maps, and being pushed into unfamiliar territory is a real disadvantage. Head-to-head win rate added secondary signal. Overall HLTV Rating and KPR, while positively correlated with winning, were the weakest contributors — likely because at this level, both teams competing in any given match are already highly rated.

The ~56% accuracy we found sits within the broader pre-match prediction range documented across sports (Bunker and Thabtah, 2019). Closing this gap would require fundamentally different features: time-stamped rating snapshots that track team form in the weeks before each match, individual player statistics rather than team averages, or in-game economy data once play begins. A region-stratified analysis would also be worth pursuing, since HLTV’s event coverage skews heavily toward European and North American tournaments.

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